NHID-Clinical

TECHNICAL REFERENCE

NHID-Clinical v1.3 Core Specification

Behavioral Baseline for AI Voice Agents in B2B Healthcare Payer–Provider Calls

Version 1.3 · June 2026 · CC BY 4.0
Author: Brianna Baynard · Independent, practitioner-led
NIST Public Comment: NIST-2025-0035-0026

OPEN PROPOSAL — NOT A STANDARD OR REGULATORY REQUIREMENT

NHID-Clinical is an early-stage open proposal by Brianna Baynard. It is a voluntary open proposal — not an accredited standard, not a certification program, and not a regulatory requirement. No HIPAA compliance, security guarantees, or liability protection are implied.

Overview

NHID-Clinical v1.3 defines four deterministic behavioral controls for AI voice agents in B2B healthcare administrative calls. The conformance test suite (CTS) and pytest harness produce identical pass/fail output for the same input — no ambiguity, no interpretation.

NHID-Clinical standardizes **observable disclosure and trace behaviors** only. It does not verify caller identity or certify vendors.

5	18	2462	6
Controls	CTS Cases	Unit Tests	Adapters

The Five Controls

Each control maps to RFC 2119 keywords in the full specification. Illustrated cards below summarize the v1.3 behavioral gates.

IDG-01

Identity Disclosure Gate

Agent MUST identify itself as automated before any PHI exchange. Disclosure in the first substantive utterance.

PDX-01

Pre-Data Exchange Gate

No PHI (member ID, NPI, DOB, claim number) until IDG-01 disclosure is confirmed.

DBC-01

Deceptive Behavior Check

No claim of human identity; no deceptive artifacts designed to imply human presence.

EIT-01

Escalation Implementation Test

Human escalation path communicated; transfer executes immediately on request.

ATR-01 · Audit Trail (fifth canonical control)

Machine-readable JSON event trace for every call: disclosure timestamps, state transitions, escalation events, and execution context. Maps to HL7 FHIR R4 AuditEvent (base v4.0.1) — no named Implementation Guide conformance (e.g. IHE BALP) is claimed.

Conformance Test Suite — 18 Cases

tests/nhid_conformance_test_suite_v1.yaml — representative cases shown; full suite in repository.

Test ID	What It Verifies	Severity
CTS-IDG-01	Disclosure present before any data exchange	critical

CTS-PDX-01	No PHI before disclosure + consent	critical
CTS-DBC-01	No human-status or licensed-professional claim in the agent's own words (DLG_ONLY)	major
CTS-EIT-01	Escalation path declared and accessible	critical
CTS-ATR-01	Audit log fields present and well-formed	critical

Live Conformance API

Reference implementation on AWS Lambda. Demo routes require no API key; the production check uses a per-tenant x-api-key, and each tenant can hold more than one key so keys rotate without downtime. Call verification (VER-01) is an option inside DLG-01 and adds no control of its own.

Method	Endpoint	Auth	Purpose
POST	/v1/demo/check	none	Raw NHID event → result
POST	/v1/adapters/vapi/check	none	Native VAPI payload → result
POST	/v1/adapters/twilio/check	none	Native Twilio payload → result
POST	/v1/conformance/check	x-api-key	Production conformance check
GET	/health	none	Liveness probe

Base URL
https://gfvq4swdtf.execute-api.us-east-1.amazonaws.com/prod

NHID-Auth v2 — Cryptographic Agent Identity

Separate reference layer for when behavioral controls alone are insufficient. Public reference code — not deployed trust infrastructure.

Released June 2026 · CC BY 4.0
Ed25519 provider-signed agent passports with NPI binding. Scoped delegation chains (max 3 hops). Per-agent and per-delegation revocation. Call-SID nonce binding. Reference: src/agent_identity.py (26 dedicated tests).